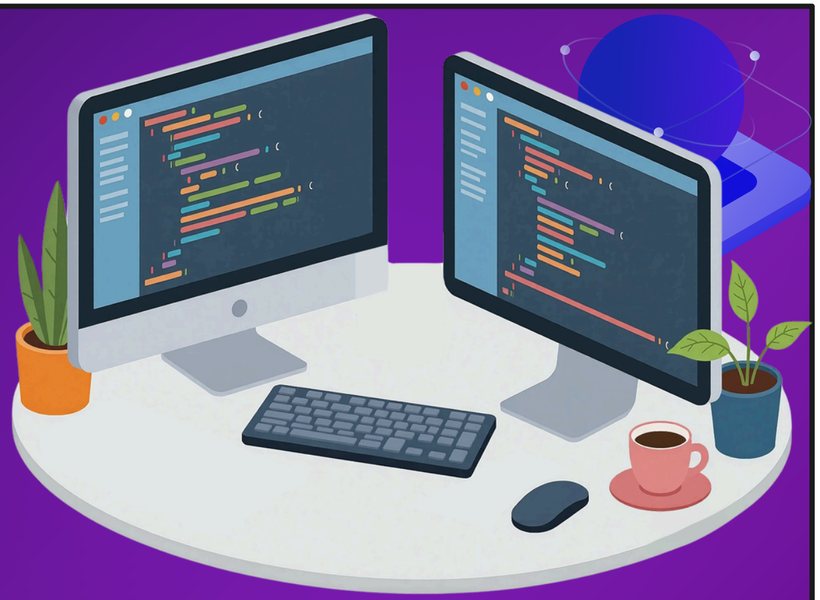
FUTURE SCHOLARS ETHIOPIA

PYTHON BASICS FOR BEGINNERS

BY

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PART FOUR

What will we learn?

- Comments
- What conditions are
- How to use if, elif, & else statements
- Using Boolean expressions and operators
- Build a simple program

Comments in Python

- Comments are notes in your code that Python ignores. They are used to explain what the code does.
- Are declared by #
- Example:

```
# This is a comment
name = input("Enter your name: ") # Get name
# Display a greeting
print("Hello", name)
```

What Is Decision-Making?

- Programs sometimes need to make decisions.
- For example:
  - If your score is 50 or higher → Pass
  - If your score is below 50 → Fail
  - If it is raining → Take an umbrella

- Python uses conditions to make these decisions.
- A condition is an expression that is either True or False
- Example

```
if condition:
    # code to run
```

The if Statement

- The if statement runs code only when a condition is True. If it is not True, it will skip it.
- Example:

```
Age = 90
if Age >= 18:
    print("You are an adult.")
```

How it works

- Python checks the condition.
- If it is True, the indented code runs.
- If it is False, the code is skipped.

Indentation

- Indentation tells Python which code belongs to the if statement.
- Example:

```
age = 20
if age >= 18:
    print("You are an adult.")
    print("You can make this decision.")
```

Using Variables in Conditions

- Boolean variables can be used directly in if statements without comparison operators.
- Example:

```
LoggedIn = True
if LoggedIn :
  print("Welcome back!")
```

The elif Statement

- The elif keyword is Python's way of saying "if the previous conditions were not True, then try this condition"

- Example:

```
A = 20
b = 20
if b > a:
    print(“a is greater than b”)
elif a == b:
    print(“a and b are equal”)
```

- In this example a is equal to b, so the first condition is not True, but the elif condition is true, so we print to screen that "a and b are equal".

Multiple Elif Statements

```
age = int(input("Input your age: "))
if age < 5:
    print("Toddler.")
elif age < 13:
    print("Child.")
elif age < 18:
    print("Teenager.")
```

- In this example, the program checks each condition in order. Then it excutes the code whose condition is True.

The else Statement

- The else keyword runs its code when the previous if/elif statements aren’t executed (or when they evaluate to False)

- Example:

```
a = 200
b = 33
if b > a:
    print("b is greater than a")
elif a == b:
    print("a and b are equal")
else:
    print("a is greater than b")
```

- In this example a is greater than b, so the first condition is not True, also the elif condition is not True, so we go to the else condition and print to screen that "a is greater than b".

Using Boolean variables

- “If age:” is equal with saying “if age == true:”
- Example:

```
age = int(input(“Please enter your age: ”))
evaluation = age >= 18
If evaluation:
    print(“You can drive”)
 else:
      print(“You are too young to drive”)
```

Using Boolean variables and operators

```
age = int(input("Enter your age: "))
activity = input("Enter your activity: ")
evaluation = avtivity.lower() == "codding"
if age >= 15 and evaluation:
    print("You can join the codding club.")
else:
    print("You can't join he codding club.")
```

Common Mistakes

1. Forgetting the colon
2. Forgetting to indent
3. Forgetting to convert input
   - Example

```
score = input("Enter your score: ")
```

Mini-project

Create a program that:

- Asks the user to enter their score.
- Checks the score.
- Displays a grade based on the score below.

90–100 –> A
80–89 –> B
70–79 –> C
50–69 –> D
Below 50 –> F